Vinod Khosla: College Degrees Are Becoming Useless | People by WTF | Episode 12

(0:30 - 1:32)

You seem like a very certain person. You take the contrarian opinion in a very certain manner. Almost certainly, with no doubt, there isn't a job where AI won't be able to do 80% of 80% of all jobs.

And that's what today's 20 or 22 year old has to take for granted. Hi Vinod, thank you for doing this. For our audience in India, which are largely entrepreneurs or going to be entrepreneurs, I don't want to introduce you.

Maybe you can tell us a little bit about your life. Your story is very well known, so I don't have to rehash it for them. But maybe you tell us a bit about life, how you began and where you are today.

(1:34 - 2:17)

Well, you know, my life has been pretty simple. I haven't done too many things, too many different things. But I was always curious about technology.

And the piece people don't understand is, or don't know, I would say, there's nothing to understand. Do you want me to stop when there's a sound overhead or keep going? Is it okay? Yeah, it's okay. Okay.

Makes it more natural. Yeah, sometimes people do and sometimes people don't. Do you do a lot of these interviews? I do a fair number.

(2:17 - 4:07)

Yeah. When I was 16, I'd go from Delhi Cantonment, where I lived, to Old Delhi to rent old magazines. That's how you got access to information.

There was no place to go to get access to magazines or others unless you could afford to buy them, and we couldn't. So, but I used to get old issues of technology magazines, especially the Electronic Engineering Times, to see what was going on. And I read about Andy Grove as a Hungarian immigrant coming and starting a company here.

And that's what fascinated me and got me interested in, why can't I start a technology company? I had very little interest and still have very little interest in business. I do it because it needs to be done, but I mostly am interested in technology and the impact it can have. So, I got enamored by this guy, and all I knew about Hungary was they had great stamps.

So, if you collected stamps, back then you always wanted Hungarian stamps, but I followed his story. Why did you want Hungarian stamps? Oh, I used to collect stamps as a kid. There wasn't like 16 things to do.

There was no internet, there was no libraries you had access to, there was not a lot of magazines. There wasn't much to read or get information from. So, you had hobbies like stamp collecting or coin collecting or I couldn't afford vintage car collecting, which some people did, but I worked on cars like that.

(4:09 - 7:30)

So, there wasn't a whole lot to do. And I used to read these magazines by renting them. And usually when the magazine got old in the area, like in the US, they got shipped and you could rent them.

So, that's what fascinated me could be done. I didn't know anybody in business. I didn't have anybody I knew who was actually in business.

So, I didn't know what that was like, neither family nor otherwise. So, you did engineering at IIT and then came to the US for your master's at Carnegie Mellon in biomedical engineering and then business in Stanford. Which part of this academic career formed you or shaped you today? You know, I would say every step was about a particular goal.

Engineering at IIT was because I was interested in technology and electrical engineering seemed pretty damn interesting. There was nothing called computer science back then. I joined IIT in 1971.

So, that was interesting. And then I got curious about biomedical engineering. There wasn't really such a discipline.

We did some projects in India in biomedical engineering, but there was no biomedical engineering department anywhere in India. And I got involved with a professor there, Professor Guha, and we did some things between IIT Delhi and All India Institute of Medical Sciences. I was curious and then I just said, well, master's in biomedical engineering would be cool.

And so, I came to the US to do my master's. But my real goal was to get to Silicon Valley and start a company. Right.

And the MBA was a necessity just to get to Silicon Valley. Right. So, I have an agenda for today, Vinod.

As an investor, the big question I have today is since everything is changing so quickly with technology changing, I want to figure out which industries, sectors might be most relevant 10 years from now, either to invest in or build a business in. The question is essentially as a young

entrepreneur in India, where do the tailwinds really lie and where should I focus my efforts on over the next decade. So, a lot of what I'm going to talk to you about today will kind of like focus on that.

And from all that I've watched in your interviews and from the few times I've spoken to you via that common pledge community we are part of, you seem like a very certain person. Certainty has been an element of most interviews I've watched, say 10 years ago, five years ago, or even today. You take the contrarian opinion in a very certain manner.

(7:31 - 8:46)

To me, I feel like certainty is not necessarily a good thing, but it's the innate human conditioning. Do you see yourself like that, as somebody who believes in something and makes up his mind and sticks to it? I would say no. I would say, well, you know this, when you built your company, mm-hmm, think about how much did you know about where you would end up? Nothing.

Nothing. That's true most of the time. Did you even know the space you were in was a valuable space to start a company in? You probably did have some hunch of that.

So, I like to say my fundamental belief in life is people have personalities and they choose to either reinforce that or go beyond that. And there's two ways. One is you don't want to fail, which fundamentally means you don't want to take risks, which fundamentally means you're not going to do anything new.

(8:47 - 9:05)

And that's a perfectly reasonable thing to do, and people get comfort in that certainty. That certainty is of very little value to me. What I like is a shot at doing the impossible.

(9:06 - 9:45)

And I like to say skeptics never did the impossible. Think about it. If you're a skeptic, you're always finding reasons why something won't work.

If you're starting your company, if you paid attention to all the things you didn't know, or didn't know you didn't know, you wouldn't start the company. And where would you be? You'd have a steady job in some corporation, and it's fine. It's a perfectly respectable thing to do, but you wouldn't be creating anything new.

(9:46 - 11:40)

Would you say a skeptic and a contrarian investor have a lot in common? A contrarian is skeptical about the incumbent norms of the day? I would say no. Skeptics, to me, are always saying why things can't be done. Entrepreneurs.

Investors are a whole different thing. I don't really think of myself as an investor. Or always saying, what if something was possible? And in the technology world, it's particularly important because I'm never certain, but I know what technology makes possible quite well.

I can make reasonable guesses at what the world could be. And then you work to make that world happen. If you happen to make it happen, it'll be successful.

Whether you're an entrepreneur or an investor. And I always think about my life that way. Like, if something's possible, I want to make it happen.

Electric cars were always possible. You knew you could put a battery and a motor in a car instead of a gasoline engine or a diesel engine. But nobody in the automotive business was going to do it.

And if they did, they made half-hearted, poor attempts. General Motors spent billions of dollars trying to do a half-assed job at an electric vehicle, EV1, EV2. Toyota did the low-risk approach of adding a hybrid engine.

(11:41 - 18:36)

You don't create the different world that way. You increment it. And there is a world of incremental innovation and lots to do there.

But people who want to be in the safe domain do that. Elon Musk created an electric vehicle, almost went bankrupt a few times in doing it. But he followed his belief there and then made the world happen.

I don't believe if we'd have electric cars dominant today, be 25% of all car sales in California today, if Elon hadn't made that world happen. So every expert in every large automotive company or every McKinsey study would say the opposite of what actually happened. And it's sort of this, if it's possible, you have to make it happen.

It doesn't matter whether you're talking about cars or rockets or the internet. You know, the telco companies, AT&T and all those people, had such naive ideas about what the internet should be. They told me in 1996, when we started Juniper, they would never adapt TCPIP.

And I do believe if Pradeep Sindhu and I hadn't started Juniper, there would be no TCPIP in the public internet. Because Cisco was not going to develop anything. And any of the traditional telco suppliers did the simple thing of listening to their customers, which is a good habit if you're trying to be incremental, a really bad thing if you're trying to be revolutionary.

For everybody who doesn't know you, Vinod, if I were to describe in like 60 seconds, I would say a very precocious, academically gifted Indian boy who went to IIT Delhi, came into the US, picked two other friends and started Sun Microsystems, and then picked a great bet after the liquidity

event of selling Sun or part of it, in picking TCPIP over the ATM protocol of the time. And then ever since then, you're more the kind of investor who's betting on edge cases, who's probably taking 100 bets, knowing 95 will fail, but the 5 will give you exponential returns. Is that a good way to describe you? I don't think so.

That's not how I think of myself. I think no matter what environment you're in. Now, what's key is I got sparked by reading a single article about Andy Grove starting Intel in Silicon Valley.

And that must have been 1970 or somewhere around then. I was 15, 16, something like that. And then sort of not stopping there, saying, how do I make this happen? I'll give you examples.

When I started at IIT Delhi, there wasn't a single computer programming class at any IIT anywhere in India. So what did I do? I said, that'd be cool to do. So we started a computer programming club and started the first real class in programming at any IIT, as a hobby club in IIT Delhi.

There was four of us and we just, we sort of made that happen. I was talking about biomedical engineering. What happened? I got curious about, as an electrical engineer, the electrical phenomena in the human body, the heart, the brain, the nervous system.

And so we started, literally started with three or four of us, the biomedical engineering program at IIT Delhi. It wasn't an official sanctioned thing. And you say to people at 16 or 17, you can make programs, new programs happen at IIT Delhi as a student or new programming happen.

It's not what people imagine possible. But I just sort of had this tendency to not accept the things as they were. So I think if I were to characterize myself, if I thought something should happen, whether it's starting a company in Silicon Valley or starting a biomedical engineering program or the first ever computer programming class in any IIT, you just sort of worked at making that happen and be clever about how to make it happen, not say, you know, the usual thing people would say is they should start a computer programming class.

But they is, who's they? Either you do it or you don't and make it happen. It's no different. That was a microcosm of what Elon Musk said.

They should be electric vehicles and I'll create them because they, which is General Motors and Ford and Volkswagen, aren't going to make that. So it's this characteristic of imagining what's possible, especially technologically, and then just making them happen. Sun was the same way.

My first startup before Sun, Daisy Systems, was the same way. And there were, you know, so you take on, imagine the possible and try and make it happen. And don't give up when it fails, even in a single journey.

And, you know, five things could have failed at any one of these efforts. Definitely at Sun or others. But never gave up.

I wanted to come to Silicon Valley. Stanford rejected me twice. Said no to admitting me.

And I always found a way to come and keep at it. So persistence, sort of passion for a vision and then persistence through being kicked in the shins many, many times. You don't give up on that.

(18:37 - 19:04)

To what end, Vinod? Because it's your passion, right? And I think the most important thing in life is a passion for a vision to make it happen. And then the persistence to make it happen as you go through the ups and downs. And frankly, also being smart about how you approach it, what teams you assemble.

(19:06 - 20:55)

Do you think that's the point of life today? Well, everybody should have a different point in life, right? If I'm a great artist, my point in life may be just great painting. And if I'm a great artist, my point may be great paintings I love. Or it could be great paintings that go for great prices at the New York art auction.

I believe those are not the genuine artists. The artist who paints for themselves is the genuine artist. It's whatever mission you set for your life.

And I think most happiness and most troubles and travails come from pursuing a mission. That's a great way to describe it. If you can tell me today at 70, Vinod, what is your passion? What are you looking forward to in life? Oh, I expect in the next 20 years at age 70, if I'm healthy, because that's the one thing nobody controls.

I expect I will affect more change in the world in the next 15-20 years than I've done in the last 50 years of my life. That's very simply the goal. And it'll be to make certain things happen that I'm passionate about.

And why is that the goal to affect change in the world? Is it posterity? No. You know, I believe when you die, you die. And I'll be buried under some tree, my ashes sprinkled somewhere or... Logically, that would mean that you have to improve the qualitative nature of life today versus worry about the change you create.

(20:55 - 21:47)

So I think the question to ask and the question I say is, let's say 25 years from now, age 95, my father lived to age 95, so that's a good metric. If I'm lying on my deathbed, what do I wish I'd spent the last 25 years on? Almost certainly it won't be I wish I made more money. Almost certainly it wouldn't be I wish I played more golf or went sailing or spent more time at resorts.

None of that I think will matter to me. Or I owned another house. Do you suspect it would be the change you brought in the world? I think it'll be how satisfied my brain is with what it was able to do and how I utilize my brain.

(21:47 - 22:06)

I think a lot of people, when asked the same question on their deathbed, they tend to say things like, I wish I spent more time with my friends. I wish I loved more. I wish I was more myself and didn't have to put on a mask or a projection.

(22:07 - 22:15)

So that's really important. So by far, I still always prioritize family over work. I know your family.

(22:15 - 22:39)

You met them, yes, at the Giving Pledge. Nobody would say I don't spend enough time there and I love spending time with my dogs. So I do prioritize that thing, but I can't impose my time on my kids.

(22:40 - 22:50)

So how much time I spend is more constrained by them, not by me. And I'm very close to all my kids. So yes, it's what matters to you.

(22:52 - 23:47)

Family definitely is right at the top. But going drinking with buddies, that's just, I don't like drinking anymore. Did you ever? No, I never did.

But you know, I didn't mind going out for a beer. I don't do that anymore. Uh, what I really enjoy is whatever time I can spend with family and helping the people I care about in any way.

But that's more constrained by their time, not by my time. And then it's use my brain to solve hard problems that nobody else will work on. So it's additionality in that sense.

If I can work on things. Look, I'm almost certain fusion, commonwealth fusion wouldn't have started. And the world needs fusion.

(23:48 - 23:59)

And it's, yes, that's a good thing to do. But it's also like selfish for me because it's a hard problem. And the hardness is what I love solving for.

(23:59 - 24:14)

Why is that, Vinod? Why do you like a harder problem? Why do people like climbing Mount Everest instead of a smaller mountain? Ego. So few can do it. So I did it.

It's a big deal. Well, so you could call it that. Yeah.

(24:16 - 24:33)

I like things that will have a large impact. So in that sense, it's legacy, but I do it in a way where I'm not building commonwealth fusion. Sam Altman's building open AI.

(24:35 - 25:04)

There's no building public transit, but the world needs all these things. So I have a document I wrote up that's a public document when I was 60. That AI will change the world.

No, no. At 60, I said, what do I want to do? By then, back then, I said next 25 years of my life. At 70, I still say next 25 years.

(25:05 - 25:15)

I feel like I'm in the same time horizon. But I wrote a document. It's a 50-page document on what do I want to work on for the rest of my life.

(25:16 - 25:38)

And I called it reinventing societal infrastructure with technology. Mostly because everybody I talked to said it can't be done. And so it was really the Mount Everest aspect of a hard problem that doesn't think that people don't think can be done that I imagined 10 years ago.

(25:39 - 25:50)

And it's a public document. It's on our website. And I just thought when I started on the writing that, I thought I'd find one problem to work on.

(25:50 - 26:23)

And I found there was no part of private GDP, non-governmental GDP that I couldn't work on in a way that would be hugely multiplicative. That means give all 8 billion people on the planet the same lifestyle that the richest 800 million people on the planet, 10% have without destroying the planet. And I've always been passionate on the climate issue because it's such a long-term problem.

(26:23 - 26:36)

People don't spend enough time on it. It's not something that can be solved when it shows up. So I defined them and started working on them.

(26:37 - 26:56)

And our portfolio 10 years later reflects all those areas pretty much. Now we do other things too because we have to have decent fund returns to keep getting the privilege of investing more. But those hard problems really challenged me.

(26:57 - 27:09)

And my old partner, John Dewar, used to say, the way to get something done is to tell we know that can't be done. And I do that. That's an ego thing.

(27:09 - 27:35)

If something can't be done, then I want to attempt it. Right. In your many successful pursuits as an entrepreneur, has the technology element of it or the ability to sell an idea element of it? What has been more valuable to you? I think I always first focus on what technology makes possible.

(27:36 - 27:52)

So one of my goals in my lifetime is to eliminate most cars in most cities. So if we put a 25-year metric on that, by 2050, we should be able to eliminate that. Now, there isn't another person in the world who told me that's reasonable.

(27:54 - 28:18)

I'm pretty sure it not only can be done, but it will be done. Will it be directly proportional to the density of population in cities? No, I think most transit should be public transit. So in a tier 3, 4, 5 town with a scarce population of say, I'm making a case for urbanization turning around, de-urbanization almost.

(28:19 - 28:33)

Say instead of millions of people crowding around one city center, you have smaller cities, fewer people, far spread out. Does public transportation work from that lens? Yes. For the following reason.

(28:34 - 28:47)

And this is what I mean by these are conceived as hard engineering, hard problems. But if you're clever, you can solve them. In fact, you're exactly right.

(28:48 - 28:56)

That's the model for India. So in the year 2000, I wrote a blog. They weren't called blogs then.

(28:57 - 29:23)

Called the Bicycle Commuting Economy for India. Mm-hmm. And essentially, I speculated instead of having everybody from the villages migrate to the major cities and having huge, huge cities like we are starting to have, there should be a city-like aggregation of people.

(29:23 - 29:35)

So you get economies of scale within a bicycle commute of the hometown or the home village of every person. Mm-hmm. All billion and a half people in India.

(29:35 - 29:45)

It was less than a billion and a half back then. And I figured there'd be 3,000 major cities. There's 700 districts in India.

(29:46 - 30:10)

So the idea that there was sort of an aggregation of a million or two million people in each side as the optimal size of a city where it's still small and personal and within a bicycle commute. But people aren't moving from Bihar to Mumbai. Mm-hmm.

(30:10 - 30:29)

And not able to go see their family very often, which leads to social isolation, broken-up families, drinking, all that other stuff. Do you think AI is bad for big cities and good for smaller cities, the model you're explaining? I think AI is good either way. Mm-hmm.

(30:29 - 30:38)

And we can come back to it. I watched your interview of the utopian and dystopian scenarios where you made a very utopian case for it. Yes, I did.

(30:39 - 30:48)

But, you know, it was around 2000 that I wrote this piece called The Bicycle Commute Economy. It might even be on our website still. Mm-hmm.

(30:48 - 30:58)

And it was specifically for how India was discussing how to set up their telecommunications network. Mm-hmm. Where to set up.

(31:01 - 31:10)

You know, they were mostly thinking landlines then, not cell phones. But my point is that is a

good aggregation. Mm-hmm.

(31:11 - 31:17)

But policy will determine that. Mm-hmm. Not technology.

(31:17 - 31:36)

Mm-hmm. And then technology will serve that scenario. I did believe with the right telecommunications infrastructure and good communications, you could enable that model to work without needing to move to a big city because a lot could be done locally.

(31:36 - 31:56)

And today, I believe that technology, even more so today, can make that possible and make a lot of the services that weren't available in a small city of a million or two million people possible today. Mm-hmm. So really have every person in India be able to live within a bicycle commute.

(31:57 - 32:02)

Now, today is a little different. But back then, most of the population was rural. Mm-hmm.

(32:02 - 32:10)

So you had to speculate that most of it will move to cities. Mm-hmm. As happened in this country and happened in other countries too.

(32:10 - 32:19)

Right. Right. China's seen some of this migration with pretty negative consequences of people moving away from their communities.

(32:19 - 32:42)

If you were a betting man, would you bet on de-urbanization or urbanization in the next decade? I think it will depend in each country on the policies they adapt and the incentive systems they set up. Which do you suspect is a better policy? Oh, de-urbanization is clearly a better policy, right? Yeah, I believe. Right.

(32:42 - 33:05)

And if I were to bring it back to our audience, Vinod, assume I'm a 22-year-old boy or girl in India. I want to learn from Vinod who has done a spectacular job of calling what industries might do well tomorrow. I want help from you in figuring out where I should spend the next 10 or 20 years of my life.

(33:05 - 33:36)

Which industry should I work in, in terms of a job, or which industry should I start a company in? So, now you come to a very hard question. You know, if I'm 20, by the time I'm 35, the year 2040, the world will be more different from today in 15 years. Well within the lifetime.

(33:36 - 33:54)

If you're going to medical school and take five, six years doing that, you waste half that time. Then the world is today from 1970. So you'd have to look back 50 years or more.

(33:54 - 34:12)

Why don't you call five years for me? I want to learn from you in making my career choice today. So if 10, 15 years are too far ahead to extrapolate, I'm 22 today, tell me for when I'm 27, five years. So the answer is the same for both.

(34:12 - 34:14)

Right. Sorry, long winded way. Right.

(34:14 - 34:34)

The point I was making is we'll see more change in the next 15 years than we've seen the last 50. And 50 years ago, the world, especially in India, was a very, very different world. You know, in 1970, I never had a TV at home or a telephone at home.

(34:34 - 34:40)

Like even when I left India, I'd never had a telephone or a TV at home. Right. Right.

(34:40 - 35:08)

It wasn't something that was in most homes in India or running water or very basic things. And that's how different the world will be in 15 years. Almost certainly, with no doubt, there isn't a job where AI won't be able to do 80% of 80% of all jobs.

(35:08 - 35:44)

So 80% of all jobs, AI will be able to do most of it within the next three to five years. So now I'm getting really in. So if you look 10 or 15 years out, there's no chance there's a job that humans do that AI can't do almost as well, with some minor exceptions, which I think today I have a hard time saying will happen for sure for regulatory or other reasons, like a heart surgeon or a brain surgeon.

(35:44 - 36:04)

But even those, AI will be able to do it better than a human, but regulation may not allow it. Right. But leaving aside those exceptions, the vast majority of jobs will be done, will be capable of being done by an AI.

(36:06 - 36:29)

And that's what today's 20 or 22-year-old has to take for granted. So what matters? The first, the single most important thing that matters is not specialization in an area. I don't learn welding or I don't learn finance.

(36:30 - 36:37)

I learn the ability to learn. Mm-hmm. Be able to move around rapidly as the world evolves and learn rapidly.

(36:38 - 37:15)

And I have to say today at age 70, I'm learning at a much faster pace than I've ever learned in my whole life, ever. And every 20-year-old should strive for this ability to jump into any new area, whether it's a physics problem or a biology problem or a finance problem, just this ability to think from first principles and learning new areas. Because yet GPT can teach you any new areas.

(37:18 - 37:32)

And AI will have that capability to be able to leverage these things well. So I would say you have to optimize your career for flexibility, not a single profession. Okay.

(37:32 - 37:54)

That's the most important advice because you don't know what will be around. Almost certainly this interview could be done without us physically meeting on real video generated by an AI. And the AI may be directed with what camera angle somebody wants.

(37:54 - 38:05)

I hear you, Vinod, when you say the rate of change will be faster. I'm going to, as a 22-year-old, be more adaptable and flexible. I still need to make a choice today.

(38:06 - 38:23)

Do I work in an IT services company? Do I work in a nuclear fusion company? Do I work in a data center? I still need to pick. Yeah. So I do think it almost doesn't matter where you start, right? Okay.

(38:24 - 38:45)

I'm saying, first, education is going to be valuable to learn how to learn, not to be good at one thing like being an accountant or a chartered accountant. Right. That's not going to be important, but that education, the process of learning how to learn.

(38:46 - 39:18)

And in fact, one of the best things about computer science isn't that you know programming. Because, you know, Google's gone from 0% AI-generated code to, I think they last claimed, 35% of all code being written in Google is being written by an AI. But it's the process of thinking that computer science teaches you really well.

(39:19 - 39:32)

And learning architectures and systems and system things, some of those things are about learning to learn. But you can do it in a number of areas. So it's not just computer science.

(39:35 - 39:56)

So I would say there's a wide range of things, but keeping your lens zoomed out for the widest angle is sort of the way to think about how you should go about your career. So you're saying, don't be a specialist, be a generalist. Be a generalist, not a specialist.

(39:56 - 40:00)

Absolutely. Right. And learn how to be a generalist.

(40:00 - 40:09)

Learn how to use AI the best. And we may be, you know, maybe it'll eliminate all jobs. AI will.

(40:09 - 40:25)

But the fact is for certain, the people who don't know how to use AI will be obsoleted by people who know how to use AI first. Right. And there, if you're dynamic and learning, then you can move with whatever's happening in the world.

(40:25 - 40:42)

I would say the following. It's just because we can't predict the future doesn't mean it won't be dramatically different. We just don't know what it will be.

(40:43 - 41:02)

I like to say something else, which is most people, most of the time, assume improbables are not

important. I've always assumed the exact opposite. Only the improbables are important.

(41:03 - 41:12)

We just don't know which of a thousand improbables is important. So what do you do in that case? You go for agility. You follow trends.

(41:12 - 41:19)

You move around. You be more adaptable and flexible. Do more first principles thinking.

(41:19 - 41:42)

It's a very different way of growing up, whether you're 20 or you're 60 or 70. And clearly I'm learning more now, as I said. And so today I feel like I can have an intelligent conversation around cell therapy in biology or about fusion.

(41:42 - 42:02)

I can do both. Or about carno efficiency in steam engines, which we discussed today for some aesthetic reason. You know, certain fundamental things, you have a broad knowledge on which you can build other knowledge.

(42:03 - 42:28)

So what is that pedestal of knowledge you acquire and how do you keep expanding it is very, very important. Most people stop learning. And I think I would say, you know, if you get a job as a waiter at a restaurant, 20 years later, you haven't gained a lot more expertise that you didn't gain in the first year as a waiter at a restaurant.

(42:28 - 42:53)

You're not exponential compounding of your capability. But other jobs, if I pick the right one, your knowledge compounds and your capability compounds over time. So I would look at the basic principle of financial compounding in knowledge and job capability as a young individual.

(42:54 - 43:22)

What will make it compound? And then you have to intersect that with what's interesting to me. What do I have passion for? If AI were to disrupt all jobs, will AI also need to be nationalized and we move back to a socialistic economy in a way? You know, look. Or does capital continue to remain the moat going into a world like that? It's very hard to tell.

(43:22 - 43:43)

It depends on policies each country adapts. And I think each country, or at least each region, will

make different choices. So why do we have capitalism? It was fundamentally for economic efficiency.

(43:45 - 44:12)

The world of Adam Smith was about economic efficiency. And the idea that if a human is doing a job, they'll be much more efficient if they're doing it for themselves and earning more income. The need for efficiency may decline dramatically in most areas.

(44:12 - 44:27)

Not in all areas. You'll still need to be efficient in the amount of steel or cement or copper you use. But efficiency's role will decline.

(44:29 - 44:49)

And hence, how you structure it and how you give the benefits of these technologies to everybody is a social choice. It's not a technology choice. It's not a choice entrepreneurs will make.

(44:50 - 45:18)

Could one argue that before Adam Smith cropped up in Scotland, if you go back many centuries before that, man was still looking out for his best, his own interest? So we might not have had the term capitalism in today's parlance. It was always about capitalism, right? In a post-AI world as well, we will feel the need to distinguish ourselves and compete. Yes, but will we have the ability to? So imagine the following.

(45:19 - 45:50)

You've seen my wife's company, CK12. If every child in India, which I hope happens in the next five years, has a free AI tutor, that's entirely possible today at a small fraction of the education budget the government has today. And it'll be better than the best education a rich person can offer their kids by hiring them personal teachers.

(45:50 - 46:07)

Almost certainly. The AI will do a better job than the most expensive tutor or teacher you can hire for your kids if you're rich and can afford anything. In that world, education is really free and it's ongoing.

(46:09 - 46:42)

And whether it's sixth grade education or college education or professional education, it doesn't matter. And you'll be able to do anything at any time and switch your majors because you don't

have to go back to college for three or five years to change from electrical engineering to mechanical engineering or from medicine to something else. I think that world is soon.

(46:45 - 46:52)

So education's free. Imagine if medical expertise was free. Every doctor was free.

(46:53 - 47:12)

Imagine every lawyer was free and every judge was free so AI could judge a legal dispute so everybody had access. Most people in India probably can't hire a lawyer to enforce their rights. And that's why we don't have real justice, access to fairness.

(47:13 - 47:47)

And of course, the courts are bottlenecked and you can have AI judges. And frankly, 10 years ago, I wrote about the poorest person can have the best wealth advisor at their level even if they're making 5,000 rupees a month because they can. Because somebody who makes a lot more won't have better advisors.

(47:47 - 48:24)

In fact, even if you wanted, just tax George and other do tax schemes and stuff, the AI will be able to figure that out better than a human. So in that world, how do you distribute wealth? If a farm worker can do the same job that an oncologist can, you pay them the same, almost certainly it'll be true 15 years from now. And I mean that almost certainly it'll be true.

(48:25 - 49:03)

So very hard to predict how this sorts out and many countries will resist it because unemployment, it causes dislocation, which people always object to. Based on that, if the AI models of today, the incumbent world are controlled by one geography, how do you think of sovereign AI in countries like India? Well, that's why we have Sarvam in India. I've always assumed countries will want sovereign AI.

(49:03 - 49:19)

That's why we invest in Sarvam in India. We invest in a company called Sakana in Japan for the same reason. Now, certain populations aren't large enough to have their own AI, but regions do.

(49:19 - 49:50)

Do you think the time is near when we all start capitalizing and protecting our data in a manner that we give access to some AI based on our preferences or geographies? Again, that'll be a function of government policy. Europeans have very different data policy than the U.S. does.

Likely outcome, you think? I don't think so.

(49:50 - 50:16)

Hmm. I think, look, you can choose not to share any information with Google today or wipe out all your information. But do you then get the benefits of the services which they can customize to you? I have no question.

(50:16 - 50:40)

I'd rather share my data and get customized services than get services that are not customized to me. I would say that if my data is the commodity that is bringing revenue to company X, I would want to get a share in the revenue in one manner or another. But you do.

(50:43 - 51:17)

Reid Hoffman has a new book called Super Agency. If you read it, he talks about the fact that, you know, if you ask people the following question, at what price would you not use Facebook? Right? How much would you pay if I said, I'm taking it away, but you can get it back if you pay for it. It's like $40 or $50 a month.

(51:18 - 51:44)

How much does Facebook make from you per month on average? About $40 or $50 a year. So you're getting, pick a number, $600 worth of value, and they're getting $50 worth of value in a year. What is a fair exchange? Everybody has the option to not buy into these services.

(51:44 - 52:07)

Now, there's lots of nuances, but is an economic surplus created? Yes. Again, it's a policy question. Government policy will have a lot more to do with these things than people realize today.

(52:09 - 52:50)

And, you know, I've argued, and I would argue, as part of Aadhaar, today we have UPI, which takes away the need for Visa and MasterCard. So it's a 3% tax on the economy that Indians don't pay, but everybody in the U.S. pays. But what if, in addition to payment services, there was medical services and education services? My bet is every Indian could get that at less than a dollar a month, or a billion dollars a year a month for each of these, or 10 billion a year.

(52:51 - 53:10)

It's so trivial in cost. If I was PM Modi, I'd be doing everything I can to get those as common services. And at that point, you know, because AI is doing the work, you don't care about

economic efficiency.

(53:11 - 53:22)

Your government is as efficient as private enterprise in these services that can be made into AI. And they should be. They're so accessible today.

(53:22 - 53:41)

They'll always be available. You know, if you live in a village, you shouldn't have to go take a day's journey to go get a basic medical checkup, and the doctor sees you for five minutes, and then you spend a day getting back to your village. It just shouldn't.

(53:42 - 54:11)

And 10 years ago in this document, I wrote about the fact, at some point in the future, much more, we are five minutes from Stanford Hospital. And I wrote about the fact that a village in India, I will get better cardiac advice, cardiac care, than I can get at Stanford, because Stanford will still in 15 years have their cardiologists, and in the village, there'll be AI. Almost certainly true.

(54:11 - 54:24)

So should the government include all of these services or as many of these services for free? They can't include everything. Housing will still be a material thing. And so you can't give it away for free.

(54:25 - 54:47)

But the software-based services that depend on expertise, personal lawyers for everybody, personal tutors for every child or adult, personal medical care advice. You know, if you need surgery or medicine, you still need a physical thing. But half the economy could be covered in almost free services from the government.

(54:48 - 54:56)

And to be honest, I'd love to do that in India. It's such a no-brainer. And make it available in the next five years.

(54:57 - 55:05)

But let's talk about what entrepreneurs have as opportunities. I get away into these. No, I heard you and Bill debate this.

(55:07 - 55:14)

Why does regulation not allow doctors? In the UK, I think you were making a case for. Yeah.

Yeah, I heard you make that debate.

(55:15 - 55:41)

Look, mental health services aren't available in India. In the UK, we have something that's used by 40% of all NHS mental health clinics to intake patients. What does intaking a patient mean? A therapist talk to them and classify their mental illness.

(55:42 - 56:03)

The AI is much more accurate than the human therapist. 93% accuracy for talk-based therapy kinds of conditions. Are you bipolar? Are you schizophrenic? Are you depressed? And AI can do the therapy.

(56:03 - 56:18)

It should be scaled infinitely in India. And it's the kind of thing people may be afraid to go or embarrassed to go ask for services. And there aren't enough therapists in India to meet everybody's needs.

(56:18 - 56:33)

But the AI can offer it for free. So in a way, you're also saying that UBI is an inevitability. I think some form of income redistribution by 2050 will be absolutely essential everywhere in the world.

(56:34 - 56:59)

Do you think it'll be income redistribution or will it be shoring up the bottom part of the population? Well, there's many ways to get there. You could give everybody income for not working. The question is, where do you get that from? There's probably another way, which is my favorite way, is to deflate the economy.

(56:59 - 57:44)

Which is what America is doing? Well, if you do excess spending, you'll get inflation, not deflation. But if a lot of services become free, you'll have a deflationary economy. So in my essay on AI dystopia or utopia, I talk about in the U.S., I believe 25 years from now, it could be that $10,000 today in 2050 buys more goods and services that you need, whether it's food or housing or medical services or entertainment or education, than $50,000 buys today.

(57:45 - 58:03)

Because the services will be provided by machines which don't deflate? Yeah, you'll have excessive production, so deflationary prices. You're no longer constrained by inputs to get outputs. In many areas, in some areas, you'll still have inputs, but you'll have much more input.

(58:03 - 58:39)

Isn't the cost of a service also dependent on purchasing power? Like how much I'm willing to pay relative to Vinod? Yeah, except if there's infinite supply. If there's an infinite supply of doctors, what do you think a doctor will cost? I think Adam Smith, like you mentioned earlier, also spoke about the marginal utility of things, wherein one agentic doctor, the first one that is created and is free, has X in value. The second, the third, the fourth, proportionately have lower and lower value to a point where they have no value.

(58:39 - 58:59)

But overnight, if I created a billion doctors for India, which is entirely feasible, what happens? They become near free. Their cost of computing or the cost of power to run a computer. Same thing happens in education.

(58:59 - 59:13)

Same thing happens in legal service. Does the cost of computer for AI ever, like your brain has a cost of compute, right? The amount of energy it takes to produce an output. Let's say that today that is far cheaper than how much AI is taking.

(59:14 - 59:48)

When do you think that crossover happens when the cost of compute in a system becomes less? I think this is a complex question because at some level, the cost will decline, but you won't take advantage in efficiency. In cost, but you'll take it in more performance. So if everybody has a super assistant, if your personal housekeeper is a PhD in every subject, that's sort of the world to imagine.

(59:48 - 1:00:28)

I know it sounds crazy, but look, I think the more interesting question for your audience is, what's one important question which you ask? How do you pick a job? We talked about that. The other question to ask is, what job should I take now? And what company can I start? And for the next 10 years, the following is true. Anybody in any industry not using AI will be obsoleted by somebody who's using AI.

(1:00:28 - 1:01:01)

So if I'm an entrepreneur in India today, I say, what's my next move? Say, where can I apply AI that obsoletes the service, either in cost or in performance or in quality that somebody's providing without AI and beat them at it? I think that's the metric for every entrepreneur. If, say, 10 entrepreneurs hear that, And there will be 10 who hear that. Yeah, 100%.

(1:01:02 - 1:01:18)

And they all use the same means to approach this problem. And AI is as democratic as it is right now. Wouldn't they all then arrive at the same solution? I think there's massive difference in the quality of entrepreneurs.

(1:01:19 - 1:01:54)

And the outcome will be a result of the quality of the entrepreneur, how they think from first principles, how strategically they think. Almost certainly, the people who only think a year out will be obsoleted by the people who think, what do I want to build in five years? But what can I sell next year? But where do I want to head in five years? Swallowing is a good analogy for entrepreneurs that I love. Imagine you want to climb Mount Everest.

(1:01:55 - 1:02:18)

And you have two people setting out. And one says, I'll get to base camp, then to camp one, camp two, camp three. I think there's camp four and then the peak, right? It's not a straight path, right? Another entrepreneur says, I'm going to get the most revenue in year two, year one and year two.

(1:02:19 - 1:02:39)

They won't go to camp one, base camp or camp one. They'll pick the highest peak that doesn't then lead to something beyond. The path to Everest through these camps is very strategically picked to have the long-term vision of getting to Everest.

(1:02:41 - 1:02:57)

Most people maximize revenue for the short term and are not thinking strategically. So the strategic entrepreneurs will do better. The entrepreneurs who pick better employees who help them think through strategy will do better.

(1:02:57 - 1:03:21)

So people selection or team building will become the most important characteristic where the AI technology is democratized if you believe it's democratized. Which investor you pick, do they have a short-term orientation or a long-term orientation? Makes a dramatic difference what kind of advice you get. Most of the advice most entrepreneurs get is bad.

(1:03:22 - 1:03:43)

So if you Google 90% of venture capitalists add no value, my name pops up because I've always said this for the last 25 years. I see such bad advice being given or short-term advice being given

by investors. So picking your investors right, picking your employees right, those make a huge difference.

(1:03:43 - 1:04:05)

When things go wrong, do you get scared and retract or pick a better strategy to survive and be more strategic? Do you think from first principle? All those things are massive differentiators. I think the shortage today is not of technology or capital. It's of great entrepreneurs who know how to make these choices.

(1:04:06 - 1:04:37)

The single most important decision an entrepreneur makes is whose advice to trust on what topic because they'll get advice from lots of people. Everybody does. Most of it is not good advice and so I feel bad for entrepreneurs who get bad advice but in the end they're responsible for deciding what advice to take and it's the most important decision they make.

(1:04:40 - 1:05:09)

Do you need an entrepreneur then if you can model for the behavior that successful entrepreneurs like you speak about a few often. If you can model for all their behavior and put it in a model and throw out outcomes like they were making the decision, why do you then need the entrepreneur? Why do you need venture capital? You know, are we there today? No. Might we be there in 10 years? Maybe.

(1:05:09 - 1:05:43)

Right. One of my speculations on the internet, on Twitter has been when will the first billion dollar company that only has 10 employees be started or has it already started? I think that's a valid question and sometime in the next 10 years we will see a company with a billion dollars of revenue that has only 10 employees. Whether it's already been started or will be started, I don't know.

(1:05:43 - 1:06:11)

Mm-hmm. So Vinod, I've been spending a lot of time with my team trying to come up with a thesis of what sectors might do better than others as this change in the world happens while it occurs. Some of the things that we've been digging data up around are things like the world will be older and sicker and people will have lesser kids than they're having or were having 10 years ago.

(1:06:12 - 1:06:38)

When people have more time and discretionary spending power they go back to their innate programming of wanting to speculate. The speculative businesses might do better. Do you have

thoughts on these or opinions like this, if you're on? On the entrepreneur side, I look at the question, who thinks from first principles? Mm-hmm.

(1:06:39 - 1:06:53)

Who's not scared of hardship? Mm-hmm. Because the entrepreneurial roller coaster has lots of highs and lows. And who has the fastest learning rate? Mm-hmm.

(1:06:53 - 1:07:04)

Right. If they're a fast learner, they'll adapt to the world as it changes. And that may be the most important characteristic of an entrepreneur.

(1:07:05 - 1:07:15)

Yeah. I'm making the assumption that you're a faster learner than everyone else here. And I want to know what you've adapted to now that I might not be seeing yet.

(1:07:16 - 1:07:44)

Well, I don't think people believe this world I'm describing. So if you look at my AI dystopia utopia paper, most of the world I'm talking about is not something people believe. But that's natural because I feel like humans are programmed to worry a lot more about fear than think of the gain that they're getting.

(1:07:44 - 1:08:01)

Twice as much according to Kahneman. But not everybody is, not all humans. And those are the people who will do well, who are optimistic about the future and then persistent through real hardship of the ups and downs to make the passion they have happen.

(1:08:02 - 1:08:19)

Right. Like we have a company that's building public transit based on self-driving where the vehicles become small and personal. You don't need a DTC bus in New Delhi to carry people.

(1:08:19 - 1:08:50)

It can be a two-person pod that drives itself from A to B. I actually have a personal question around this. For the last couple of years, I had come up, I almost made an assumption that mobility will move to electric from IC, fossil fuel based. And I bet on everything from scooter companies to car, truck, bus, even a flying taxi company.

(1:08:50 - 1:09:05)

But the adoption to EV in this last cycle seems to be slowing down. Do you agree with that thesis

that all vehicle, all mobility will be electric in nature? And if yes, why is it slowing down? On land, yes. Right.

(1:09:05 - 1:09:22)

Almost certainly. Why is it slowing down the pace of adoption? You know, look, anytime you're an incumbent, you want to slow things down. It's to your advantage that things change slowly because you adapt more slowly.

(1:09:23 - 1:09:29)

And so incumbents always do that. And there's always fits and starts. It's never a straight line.

(1:09:30 - 1:09:49)

But I can't see, you know, some of the U.S. companies are slowing down. But what's happening? BYD in China is the fastest growing company in the automotive world. Actually, even in India, a lot of companies are slowing down.

(1:09:50 - 1:10:00)

Yeah, and that may be, but that doesn't change eventually where it'll get to. Yeah, I hope so too. But yeah, I'm pretty certain that'll happen.

(1:10:02 - 1:10:26)

But the more the uncertainty, the harder it is to pick the right company to invest in. Do you think energy and data centers as a business, sufficient legroom and tailwinds to be, if I were 22 and starting off, to spend a decade in them? By energy, I don't mean the means of generating energy. It could be fusion.

(1:10:27 - 1:10:37)

It could be something else. But focusing on energy. You know, I would say there's some combination of where your passion is and expertise.

(1:10:37 - 1:10:55)

Passion counts for more than most people think, right? Because you persist through ups and downs. And if you don't have passion, you give up. It's like you don't give up on your religion just because you had a hard time.

(1:10:56 - 1:11:18)

So, someone wise once said, a emotion, which is a passion, ceases to be a passion once you

develop a clear understanding of it. So you're not really, it's very hard to be passionate about something that you've understood. Religion could fall in that territory.

(1:11:18 - 1:11:48)

Nobody can truly understand it. And I've heard you in interviews say that passion is not necessarily the pivotal force that should define which sector you should work in. So, I think in the world of the past, I used to tell kids when the advice you get is follow your passion, I would say, follow your passion if you're willing to starve and not make a living for your family.

(1:11:48 - 1:12:03)

Live on a beach in a hut and stuff. I remember you saying this. So you can go surfing and you can absolutely go live in Hawaii and survive, but you're not paying for a Stanford education for your kids if you're surfing.

(1:12:04 - 1:12:37)

So, you have to be realistic about, it may still be a good trade-off, but everybody has to be explicit on what trade-off they're making. If you bet your life on being the best cricketer, you don't make it, then you're a waiter. Right? Are you willing to take a 0.001 probability you'll be a world-class cricketer on the Indian cricket team or in the rest of the probability you're a waiter? That's a choice you get to make.

(1:12:38 - 1:12:57)

That used to be true. I think we will go in a world where you can pursue much more of your passion. I think that will change and that is where AI will free humanity to pursue what they're interested in and to change your passion if it changes over time.

(1:12:58 - 1:13:10)

My passions aren't the same from when I was 20. Yeah, fair. In all that you're saying, Vinod, you're coming across very optimistic and utopian about this whole thing.

(1:13:10 - 1:13:38)

I know you don't believe in bubbles per se, but as an investor, you have seen the many cycles, many more than I have. Do you think we could be in a bubble today and overestimating, I'm not saying overestimating the change, but the pace or the rate of change? I would make the following differentiation. Investment is a different thing than building.

(1:13:40 - 1:14:14)

So when you're an investor, by and large, you're competing with other investors that have only

two emotions, fear and greed. And so every time there's an opportunity-rich space, you get a bubble. If you look at the dot-com bubble in 1998 or so, if you look at stock prices, they went up and then they collapsed and then they normalized.

(1:14:16 - 1:14:49)

But I don't look at that world because I don't think of myself as an investor. I say, what happened to internet traffic? You can't tell where the bubble is by looking at internet traffic. One of the original bubbles was in the 1830s in England when railroads first came and you got the right to build a railroad between two cities, you could offer scripts, essentially do a stock, an IPO.

(1:14:52 - 1:15:14)

And of course, there was a huge bubble in script prices and then a collapse. The same thing, if you look at the miles of railroad built, you can't tell when the bubble was in the 1830s. So what is reality for you? To me, the reality is internet traffic, miles of railroad built, AI applications.

(1:15:14 - 1:15:28)

AI valuations isn't something I worry about, but there surely will be a bubble. Almost certainly. Will be? Will be or we are in one.

(1:15:28 - 1:15:39)

Right. I like to say, even in 2025, most investments that are made will lose money. Right.

(1:15:39 - 1:15:56)

You believe that? The vast majority, I might even guess north of 80%. But more money will be made than lost. So the few things that succeed are playing in such large markets.

(1:15:56 - 1:16:23)

It's the same issue of there's 10 startups, one will make more money than the other nine lose in every area. Of the next thousand startups, 990 can lose money and you can still make money by investing in the right portfolio. So I do think we will see that.

(1:16:23 - 1:16:29)

I do think we will see a collapse in valuations. Most likely. You can't never be certain.

(1:16:30 - 1:16:47)

But it's such a large market we are playing in, such a large global transformation that there will be many more opportunities than we've seen ever before. And that's good news for entrepreneurs. And almost certainly most of this won't be done by large companies.

(1:16:48 - 1:17:15)

Do you have a view on crypto and blockchains? They're not allowed in India. It's a world where we have seen sudden adoption in the West and some neighbors of ours, but... Again, as you've seen, crypto has been a function of politics, not of reality. You don't think? I think there's very different worlds of crypto.

(1:17:16 - 1:17:30)

There's cryptocurrency, very, very different than the blockchain. The blockchain has a lot of uses. And if you said to me, will you see adoption of stablecoins? Absolutely.

(1:17:30 - 1:17:40)

It makes total sense to adapt stablecoins. Okay? Every economy should have a stablecoin. The Indian government should issue a stablecoin.

(1:17:40 - 1:18:04)

And then great place to do it with the Aadhaar system. But don't you think something is wrong with the premise that you didn't trust regulation in the formal economy? So you created a blockchain outside and a crypto on top of it. Now, if you denominate the stablecoin with the same dollar or the same currency, aren't you going around in a circle? Here's the problem.

(1:18:06 - 1:18:30)

When you don't have trust, when is the blockchain important? When you don't have centralized trust. I don't think distributed trust works across borders in all areas altogether. So the vast majority of uses of crypto are illegal stuff.

(1:18:31 - 1:18:47)

Gun trades, drug trades, sex trafficking, arms trades. It's the vast use of crypto. Or people in Argentina trying to get their money out of the country or in China, people trying to get their money out of the country.

(1:18:48 - 1:19:03)

So in some places, they're doing exactly what the global economy would want them to do, which the government doesn't want them to do. But it's mostly illegal. And then there's speculation.

(1:19:04 - 1:19:17)

And of course, related to that, fraud. Are there good uses for the blockchain of distributed trust?

Yes. We have a company building a cell phone network based on distributed trust.

(1:19:18 - 1:19:37)

Everybody can contribute their own access point. Now, there's another company building Google Maps competitor based on how much you contribute to mapping. Like, there's lots of good uses for the stablecoin, but there's lots of bad uses.

(1:19:37 - 1:19:56)

The problem is the bad uses prevent the good uses. And so stablecoins' currency mechanism is better than speculation. And from that lens, the banking industry will likely get disrupted sooner than later.

(1:19:56 - 1:20:03)

With stablecoins, it would be, yes. Yeah. The cost of transaction and the... Yeah.

(1:20:03 - 1:20:25)

And UPI has started to do that in India, but it should be much, much cheaper. And, you know, look, you don't need to have the whole legal system. You can have software contracts for at least half the things for which you have a legal system around contracts, right? So software contracts are a really good use of the blockchain.

(1:20:25 - 1:20:35)

Mm-hmm. Automatic settlement of the contracts financially, really good use of the blockchains. So I hope those come up.

(1:20:36 - 1:20:56)

Specifically to your home country, Vinod, of India, which has a disproportionately high number of IT services being exported. I can't remember the number, but somewhere around 200 billion. What do we do? How do we become resilient? I think it's very, you know, BPO as a business will disappear.

(1:20:57 - 1:21:04)

Yeah. Software IT services will mostly disappear. Disappear means transform pretty radically.

(1:21:05 - 1:21:23)

Whether some of those companies can transform or not will determine whether they survive or not. What do you suspect? Will they? Will the incumbents transform? Today, I see them adjusting incrementally, not radically. Mm-hmm.

(1:21:24 - 1:21:35)

Here's the problem, right? If I can do a system integration service for one-fifth the cost, the customer will always take one-fifth. Mm-hmm. Modulo trust.

(1:21:38 - 1:21:54)

So will some of these vendors drop their prices by 5x or 80% or at least by 60% and then expand their services? Hard to tell. I do apologize. I'm late for a dinner.

(1:21:54 - 1:21:56)

Yeah, no, thank you. Thank you for doing this, Vinod. This was fun.

(1:21:56 - 1:22:00)

Thank you for having done it. Yeah, this was fun. Well, I'm happy to catch up sometime.

(1:22:01 - 1:22:02)

Yeah, 100%. We'll do it.